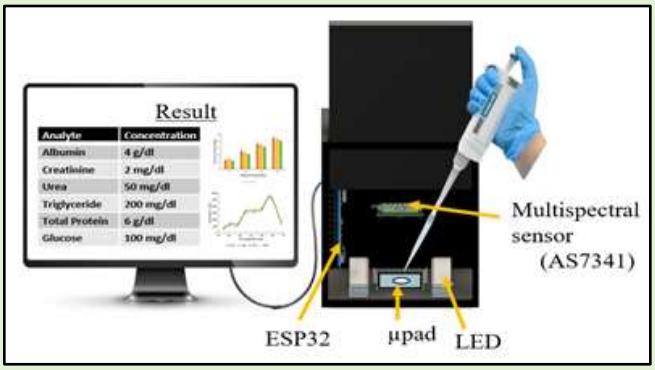
Microfluidic Paper-Based Portable Biochemistry Analyzer for Blood Parameter Sensing

Sangeeta Palekar*, Bidipta Rana, Ayushh Khaire, Abhijit Awasthi, Minakshi Mahule, Jayu Kalambe*

***Abstract*— A compact, portable, paper-based biochemistry analyzer has been developed for cost-effective, point-of-care blood parameter analysis. The device employs wax-printed paper-based microchannels (µpads) coated with specific reagents to detect critical blood parameters. It analyzes blood serum for sugar, proteins, enzymes, electrolytes, and lipids, aiding in diagnosing, monitoring, and treating various medical conditions. A 3D printed device is designed that houses a multi-spectral sensor (AS7341) interfaced with an ESP32 microcontroller measures absorbed light to analyze concentration in real time. To validate the performance of the developed device, various parameters including albumin, creatinine, urea, triglycerides, total protein, and glucose are tested. The µpads allow precise detection using a minimal sample volume of 100 µL, making it especially suitable for low-resource settings. The device's performance showed high accuracy, with correlation coefficients greater than 0.98 for all analytes. Additionally, the linear range for respective analytes is: albumin up to 8 g/dl, creatinine up to 20 mg/dl, triglycerides up to 1000 mg/dl, total protein up to 15 g/dl, and glucose up to 600 mg/dl, with corresponding limits of detection (LOD) at 0.134 g/dl, 0.087 mg/dl, 0.055 mg/dl, 0.094 g/dl, and 0.015 mg/dl, respectively. The portable design, combined with its ease of use and low-cost components, makes this analyzer a practical solution for point-of-care diagnostics, particularly in remote or underserved regions where conventional laboratory equipment is impractical due to size, cost, and operational requirements. By overcoming the limitations of traditional bulky analyzers, this device has the potential to significantly enhance accessibility to biochemical diagnostics in healthcare.**



***Index Terms*— Point-of-care diagnostics, paper-based microfluidics, blood parameter analysis, portable biochemistry analyzer, ESP32, multi-spectral sensor, statistical accuracy.**

I. INTRODUCTION

Blood is the human body fluid has been the primary indicator of an individual's health which is being widely studied over the decades. It is an extremely complex fluid that is made up of different substances, such as proteins, many chemical compounds, and cell and their components.

Laboratory-based blood analysis is very crucial and follows a well-established process, related to the concentrations of important blood components and helps them make diagnoses and treatment decisions. It is usually carried out in medical or laboratory environments, which emphasizes accuracy and measurement automation [1]

To ensure comprehensive blood analysis, it is crucial to detect and evaluate all substances present. Various techniques have been developed for analyzing key blood parameters, including glucose levels, albumin, triglycerides, and total protein, among others. Some of the commonly used techniques include spectrophotometry, chromatography, electrophoresis, and enzyme-linked immunosorbent assay (ELISA). These methods enable accurate monitoring of these vital components, contributing to effective diagnosis and treatment.

The extensive applications of enzyme-based biosensors in biotechnology, such as environmental control, immunoassays, and clinical monitoring, have continuously gained attention. This approach uses enzyme-based catalysis to enable the sensitive and selective detection of target molecules [2]. However, enzyme-based biosensors can suffer from degradation over time, particularly in harsh conditions, reducing their stability and shelf life, which can be problematic for long-term applications.

Immunoassays in enzyme-based biosensors are indeed powerful due to their ability to leverage the high specificity of antibodies. This allows them to detect even low concentrations of proteins or peptides with precision. Their sensitivity and linearity make them particularly useful for applications where accurate quantification of biomarkers is critical [3].

High-performance liquid chromatography (HPLC) is widely employed for the separation and analysis of numerous substances. It is favored for its rapid experimental time, requiring only small sample amounts, while providing high

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resolution and excellent reproducibility, making it an essential tool in various analytical applications[4]. However, HPLC can be expensive due to the cost of solvents, columns, and equipment, and it requires skilled personnel to operate and interpret results.

Mass spectrometry (MS) is a highly specific analytical technique, well-suited for clinical laboratories. It works by detecting analyte ions in the gas phase. The method's key strength lies in its ability to provide exceptionally specific detection, making it invaluable in various analytical applications[5]. Despite its specificity, MS is expensive and requires specialized instruments and expertise. Moreover, improper sample preparation can lead to contamination or loss, affecting accuracy.

A modified version of two-dimensional (2-D) polyacrylamide gel electrophoresis allows for the consistent detection of differences between two protein samples using just a single gel. Two-dimensional polyacrylamide gel electrophoresis (2-DE) is now the main method for analyzing complex protein mixtures because of its superior resolution and sensitivity [6]. However, 2-DE is labor-intensive, requires high levels of technical skill, and is prone to variations in gel quality, which can compromise reproducibility.

The biochemistry analyzer is a device that measures and examines different biological and chemical components of human blood, urine, and other bodily fluids [7]. There are various types of existing biochemistry analyzer, automated biochemistry analyzer, point-of-care analyzer, the image acquisition biochemistry analyzer. Automated biochemistry analyzer is liquid based system automatically pumps the sample and reagent, incubates the volume to be measured, and measures it. The major disadvantage of this system is volume requirement of the sample which is about 0.5 mm. The point-of-care analyzer is either liquid or dry test. In case of liquid test, sample need to be added manually and thus centrifugation and measurement takes place. In dry test, point-of-care analyzer use test-strips which is layered with reagent to measure specific parameter. The performance is affected if the test-pads are not properly stored. The image acquisition biochemistry analyzer uses USB mini camera that is placed above the cuvette container and measures the desired parameters. This requires cuvette, spectrophotometer , USB camera module and a proper set-up measure the blood parameters [8]

Population health monitoring requires biological fluid analysis [9], but these measurements are challenging to carry out in remote areas, like those found in less developed nations, during emergencies, or in home health care settings [10] Traditional laboratory equipment can quantify biological samples, but because of their size, cost, need for specialized workers, and amount of biological samples, it is not appropriate to use it in these circumstances [11].

To overcome the above-mentioned disadvantages, this paper presents portable biochemistry analyzer which is basically a paper-based clinical biochemistry analyzer. This consists of wax-printed paper-based disposable microchannel-pads (µpads), reagent coated over the µpads and stored, multi-spectral sensor interfaced with micro-controller which senses the intensity of different blood parameter and ultimately gives the corresponding concentration of the blood parameter, making it a portable and low- cost biochemistry blood analyzer.

II. EXPERIMENTAL

A. Reagents, Chemicals, and Components used:

In this experimental setup, various reagents, chemicals, and components are carefully selected to ensure the accurate detection and quantification of blood parameters such as glucose, proteins, enzymes, electrolytes, and lipids. These materials play a critical role in facilitating the chemical reactions necessary for measurement, as well as in ensuring precise data acquisition and processing.

Using the Bromocresol Green (BCG) method, albumin samples are tested. For in-vitro diagnostic purposes, the albumin test reagent kit is used to measure the amount of albumin in serum or plasma. When serum albumin is exposed to acidic conditions, such as when BCG is present, it forms a green-colored complex. The R1 BCG Reagent (Blank) is prepared by combining Succinate Buffer (90 mmol/L) to maintain pH and Bromocresol Green (0.26 mmol/L) to create a green complex [12]. The creatinine test reagent kit was used to measure creatinine content in serum/plasma or urine using Jaffe's Method. The R1 Creatinine MR Reagent was prepared by combining picric acid, sodium hydroxide, EDTA, and surfactants, forming a yellow to orange Creatinine Picrate complex[13].

For urea, the Glutamate Dehydrogenase (GLDH) method was used to estimate urea or blood urea nitrogen (BUN) in serum/plasma. The urea test reagent uses a combination of R1 Urea BUN Reagent and R2 Enzyme Reagent. The R1 Urea BUN Reagent contains TRIS buffer, ketoglutarate, urease, and glutamate dehydrogenase, while the R2 Enzyme Reagent consists of NADH. This process results in the formation of glutamate as the final product. The total protein test reagent kit is utilized to measure the total protein in serum or plasma using the Biuret method. The R1 Biuret Reagent is created by combining copper sulfate (20 mmol/L), sodium potassium tartrate (15 mmol/L), and sodium hydroxide (30 mmol/L), resulting in the formation of a violet-colored complex.

The triglyceride test reagent kit is a medical device designed to measure triglycerides in serum or plasma by GPO-PAP method. The R1 Triglyceride reagent is prepared by combining lipase, glycerol kinase, glycerol phosphate oxidase, peroxidase, and a buffer, resulting in the formation of a purple quinoneimine dye. The glucose test reagent kit is designed to measure glucose levels in serum, plasma, blood, or bodily fluids using the GOD-POD method. The reagent is created by combining Glucose Oxidase (GOD), Peroxidase (POD), phenol, 4-Amino Antipyrine, buffer, stabilizers, and activators, resulting in the formation of a red quinoneimine dye.

The materials and components used for detecting blood parameters include a specially designed 3D-printed black enclosure, made from PLA material, with dimensions of 6 cm x 6 cm x 6 cm. This enclosure houses a sensing platform where test samples are analyzed. For consistent illumination, two LED strips are placed inside the enclosure. Central to the detection system is a multi-spectral light sensor (AS7341), which is interfaced with an ESP32 microcontroller for precise data acquisition and processing. Additionally, grade 1 Whatman filter paper and a parafilm layer are used in fabricating µpads, allowing for the application and handling of blood samples

within the device. This setup ensures accurate and reliable measurement of various blood parameters.

B. Fabrication of µpads for blood parameter sensing

The fabrication of microfluidic paper-based analytical devices (µpads) involves creating a dual-region structure on a circular pattern, comprising an inner hydrophilic zone surrounded by an outer hydrophobic zone (Fig 1) [14]. This design is crucial for the controlled movement of fluids within the device, enabling specific reactions and tests to be conducted in designated areas. To create the mold for the µpads, standard A4 copier paper was used. Circular structure with a diameter of 1.8 mm were precisely cut at equal intervals using laser cutter on the paper to ensure uniformity in the final design. This step was critical for maintaining consistent spacing and structure within the device, which is essential for reliable fluid dynamics. Once the circles were cut, the mold was dipped into molten wax for 10 seconds. The commercially available wax was heated to a suitable temperature, ensuring it was fully melted and able to coat the mold uniformly. After dipping, the mold was allowed to dry at room temperature. This drying process solidified the wax, forming a stable coating on the mold that would later be transferred to the filter paper.

The wax-coated mold was then placed between two sheets of Whatman #1 filter paper. This filter paper was chosen for its high purity and consistent quality, making it ideal for microfluidic applications. The sandwich structure, with the mold between the filter papers, was passed through a hot laminator. The lamination process was a critical step, as it involved melting the wax onto both sides of the filter paper, creating a hydrophobic barrier that defines the outer zones of the µpad. The entire lamination process was carried out in a single pass to prevent excessive spreading of the wax, which could compromise the precision of the hydrophobic barriers. An optimized temperature of 120°C was selected for this process. This temperature was crucial in ensuring that the wax melted sufficiently to imprint the mold's design onto the filter paper without causing unwanted spreading that could blur the boundaries between the hydrophilic and hydrophobic zones. In addition to the wax, a parafilm layer was coated on one side of the filter paper. This layer further reinforced the hydrophobic barrier, preventing the filter paper from absorbing liquids and ensuring that the hydrophilic and hydrophobic zones functioned as intended. The parafilm coating was essential in maintaining the integrity of the device, especially during fluid handling.

The effectiveness of the fabricated µpad was validated by introducing dyed water to the device. The water's movement was carefully observed to ensure that it stayed within the hydrophilic zones and was effectively contained by the hydrophobic barriers. The absence of leakage or disruption in the water flow confirmed that the wax barriers were well-formed and that the device would function reliably in practical applications. This validation step was crucial in confirming the quality and functionality of the µpad, ensuring it would perform as needed in analytical tasks.

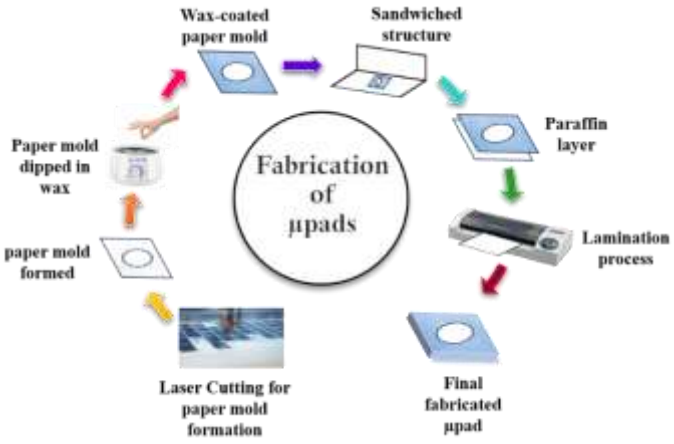


Fig. 1: Process of fabrication of µpads

C. Design of paper-based Portable Biochemistry Analyzer

The developed portable biochemistry analyzer combines Beer-Lambert's law with advanced multi-spectral sensing technology to measure light intensity and determine the concentration of various blood parameters as shown in Fig. 2. Encased in a compact, PLA-based enclosure, the device is designed for easy transport and use in clinical and field applications, including remote locations without access to traditional laboratory facilities. By integrating multi-spectral sensing technology with the proven principles of Beer-Lambert's law, this portable biochemistry analyzer offers an accurate, reliable, and cost-effective solution for measuring blood parameters. Its compact, user-friendly design makes it suitable for use in clinics, remote locations, and emergency settings, offering versatile and efficient healthcare monitoring. Its portability and ease of use make it an ideal solution for point-of-care testing, remote healthcare, and emergency medical situations.

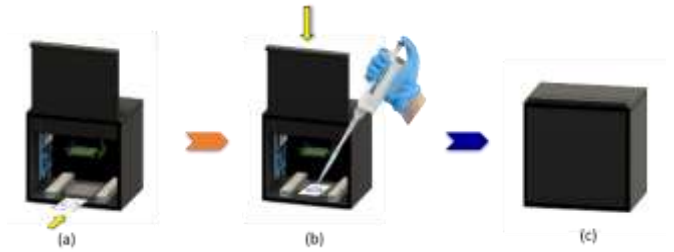


Fig. 2: (a) The fabricated µpad is positioned within the sensing platform, (b) Prepared sample of 100µl is deposited over the µpads, (c) The lid is closed to create a controlled dark environment, optimizing conditions for the AS7341 sensor to obtain accurate measurements.

A precisely designed sensor holder at the top securely positions the AS7341 multi-spectral sensor, ensuring accurate alignment with the blood sample. This design reduces measurement errors and enhances overall accuracy. The AS7341 sensor, equipped with 11 channels, can detect light and color across a wavelength range of 415–680 nm. This multi-wavelength capability is crucial for accurate detection, as various blood parameters respond to specific wavelengths.

At the bottom of the device, the sensing platform holds microfluidic pads (µpads) where blood samples are deposited. These µpads are aligned with the sensor holder for optimal detection, as shown in Fig. 2(a). The platform's design requires only small sample volumes (e.g., 100µL), minimizing reagent use and reducing costs, making the device ideal for low-resource environments. Once the sample is applied to the µpad (Fig. 2b), it reacts with specific reagents that produce a color change. This change correlates with the concentration of the targeted blood parameter.

The AS7341 sensor then detects the absorbance of light at various wavelengths. Consistent illumination from two internal LED strips ensures stable lighting conditions during measurements (Fig. 2c). The device uses an ESP32 microcontroller, which connects to the AS7341 via the I²C protocol to collect spectral data. The microcontroller processes this data, analyzing the raw absorbance readings using algorithms to calculate the concentration of substances like glucose, proteins, enzymes, electrolytes, and lipids. By applying Beer-Lambert's law, which relates absorbance to concentration, the device can accurately quantify these parameters. The device features a user-friendly interface that simplifies the blood analysis process, making it accessible even to users with minimal technical expertise. This interface guides the user through sample application, measurement, and result interpretation, ensuring a quick and straightforward process. The portability and ease of use make this device ideal for point-of-care testing, remote healthcare, and emergency medical situations.

D. Sample Preparation:

Sample preparation is a critical step for ensuring accurate results in the paper-based portable biochemistry analyzer. The process involves using blank, reagent, and standard solutions to create samples of varying concentrations for different analytes. Given that the analyzer requires only 100 microliters per test, small volumes are prepared for analytes such as albumin, creatinine, urea, triglycerides, total protein, and glucose. Each analyte is incubated for a specific time to allow proper reaction with the reagents. Blank solutions are used for baseline measurements, and all samples are carefully handled to avoid contamination. This approach ensures precise calibration and consistent, reliable results across different testing environments. The details are summarized in the Table I below.

TABLE I
SAMPLE PREPARATION

Analyte	Sample Volume	Reagent Volume	Incubation time	Standard Concentration
Albumin	1µl	100µl	5mins	4g/dl
Creatinine	10µl	100µl	--	2mg/dl
Urea	1µl	100µl	--	50mg/dl
Total Protein	1µl	100µl	5mins	6g/dl
Triglyceride	1µl	100µl	10mins	200mg/dl
Glucose	1µl	100µl	10mins	100mg/dl

III. RESULTS AND DISCUSSIONS

A. Optimization of µpads

To determine the optimal diameter for the micro-pads (µpads) used in the developed portable biochemistry analyzer, various experiments were conducted to evaluate performance across different sizes. The experiments focused on measuring key metrics such as accuracy, reliability, and sample utilization for three different µpad diameters: 1 cm, 1.5 cm, and 1.8 cm. Each diameter was tested for its ability to ensure precise sample detection and even distribution of a 100 µL sample across the pad surface.

Results from these experiments demonstrated that the 1.8 cm µpad consistently provided the most accurate and reliable measurements, with lower variability and higher repeatability as illustrated in Fig. 3. The larger surface area of the 1.8 cm pad minimized the effects of minor placement errors, allowing for more consistent readings. Additionally, the sample spread evenly without gaps or overflow, reducing reagent waste and optimizing sample use.

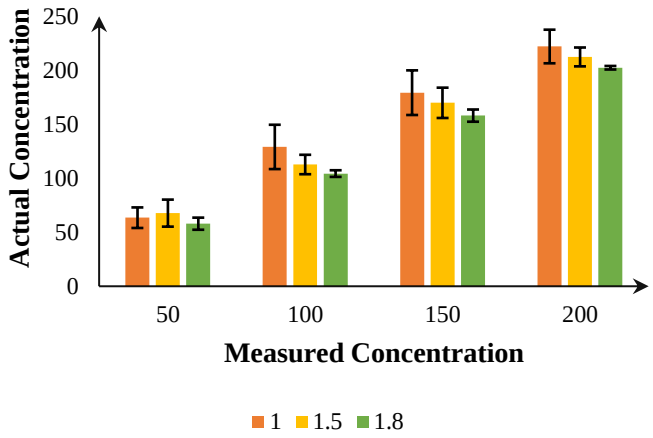


Fig. 3: Experimentation performed for different diameter of µpads for optimal utilization and sensing of sample

B. Chemical Compatibility against wax-barrier:

To confirm the chemical compatibility of the fabricated wax patterns, various solvents, including creatinine, urea, albumin, triglyceride, total protein, glucose was tested. Figure 4 illustrates the stability of these wax patterns after exposure to these solutions. The analysis demonstrated that nearly all the chemicals maintained their stability within the defined regions, indicating robust chemical resistance [15].

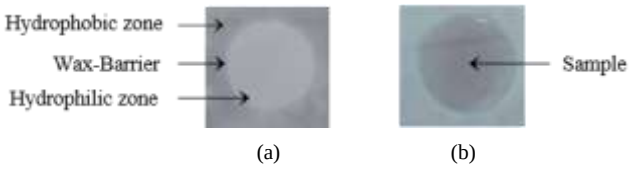


Fig. 4: (a) Fabricated µpad, (b) Wax barrier restricts the dispersal of the chemical

An important aspect of this study was the distinction between the hydrophilic and hydrophobic regions of the wax patterns. The hydrophobic regions play a critical role in preventing chemical overflow by repelling the solutions, ensuring that they

remain confined within the hydrophilic areas designed to hold the samples. This selective interaction between the different regions is essential for accurate measurements, as it prevents cross-contamination and ensures that the chemicals are precisely contained within their designated areas. This design feature enhances both the reliability and precision of the overall system, supporting its effective use in various applications.

C. Absorbance spectra for Wavelength Selection

In this work, the multi-spectral sensor (AS7341) with 8 wavelength filters is used to determine the optimal wavelength for analyzing different analytes. Each filter corresponds to a specific range of wavelengths, allowing the sensor to capture absorbance readings across the spectrum.

To find the most suitable wavelength for analyte detection, different concentrations of the analyte samples are prepared, and absorbance readings are taken at each of the 8 wavelength filters. The filter showing the highest absorbance is identified as the most appropriate wavelength for the analyte, as it represents the peak absorbance where the analyte interacts most strongly with the light. This wavelength is then used to calculate the final concentration of the analyte in the sample, providing the most accurate and sensitive measurements for the experiment.

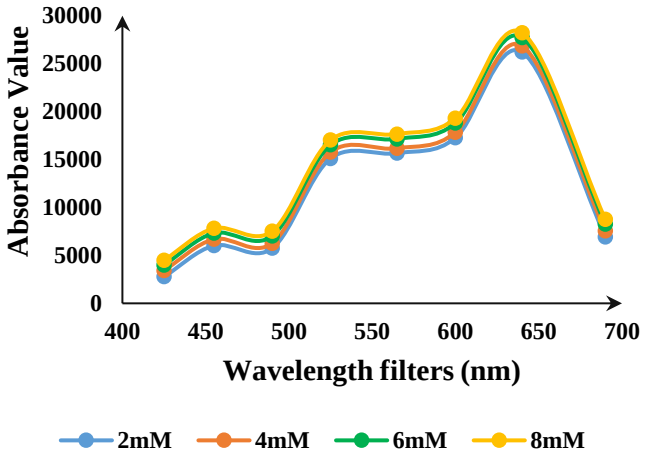


Fig. 5: Absorbance measurements were recorded across all available wavelengths of the AS7341 sensor to identify the most suitable wavelength for detecting a specific analyte.

We demonstrated this approach for albumin and determined the suitable wavelength to be 630 nm. Similarly, after analyzing the results for creatinine, urea, triglyceride, total protein, and glucose, we identified the optimal wavelengths as 520 nm, 650 nm, 540 nm, 570 nm, and 505 nm, respectively. This method ensures that the wavelength chosen is specific to each analyte, maximizing the precision of concentration calculations based on absorbance data.

C. Estimation of concentration of various blood parameters

In clinical biochemistry, accurately measuring the concentration of analytes in biological fluids is essential for effective diagnosis and treatment. A fundamental principle underlying these measurements is the Beer-Lambert Law, which relates absorbance to concentration. Absorbance (A) is calculated using the formula:

$$A = \log_{10}(I_o / I)$$

Where:

- I_o is the spectral intensity of the incident light (before passing through the sample).
- I is the spectral intensity of the transmitted light (after passing through the sample).

The spectral intensity values are captured by the AS7341 sensor, a spectral sensor that detects different wavelengths of light.

Two commonly used techniques for sensing blood biochemistries are the End-Point Method and the Two-Point Kinetic Method. The End-Point Method is a widely used approach that measures the absorbance of a sample after a biochemical reaction has reached completion. In this method, a reagent is added to the sample, triggering a chemical reaction that produces a colored complex. The absorbance is then measured at a specific wavelength corresponding to the peak absorbance of the colored product. This method provides a clear representation of the reaction's outcome, allowing for the quantification of analytes based on the absorbance value obtained at the end of the reaction. The concentration can be determined using the equation:

$$Analyte\ Concentration = \frac{A_s - A_B}{A_{std} - A_B} * (Std\ Conc.)$$

Where: Absorbance of Sample (A_s): Absorbance measured for the test sample. Absorbance of Blank (A_B): Absorbance measured for the blank solution. Absorbance of Standard (A_{std}): Absorbance measured for the standard solution.

This formula is based on the principle that the absorbance of a solution is directly proportional to the concentration of glucose in the sample. The absorbance difference between the sample and blank, compared to that of the standard, is used to calculate the concentration in mg/dl or g/dl.

In contrast, the two-point kinetic method measures the rate of reaction over time, providing insights into the kinetics of the biochemical reaction. This method is particularly important for analytes where the reaction progresses continuously and may not reach a stable endpoint within a short time frame.

The procedure typically involves measuring the absorbance at two intervals—often set at the beginning (30 seconds) and later (90 seconds) of the reaction. The absorbance values are then used to calculate the concentration of the analyte based on the change in absorbance over time. The concentration is calculated using the formula:

$$Analyte\ Concentgration = \frac{(A_{t1} - A_{t2})}{(A_{s1} - A_{s2})} * (Std.Conc.)$$

Where:

- A_{t1} and A_{t2} are the absorbance values of the test sample at 30 seconds and 90 seconds, respectively.
- A_{s1} and A_{s2} are the absorbance values of the standard at 30 seconds and 90 seconds, respectively.
- Std. Conc. is the known concentration of the standard solution.

This method provides a more dynamic approach to quantifying analytes and is especially useful for substances like serum creatinine, where reaction kinetics play a crucial role. By monitoring the rate of change in absorbance, the Two-Point Kinetic Method can reveal information about the reaction's progress, making it a valuable tool in situations where the reaction time varies.

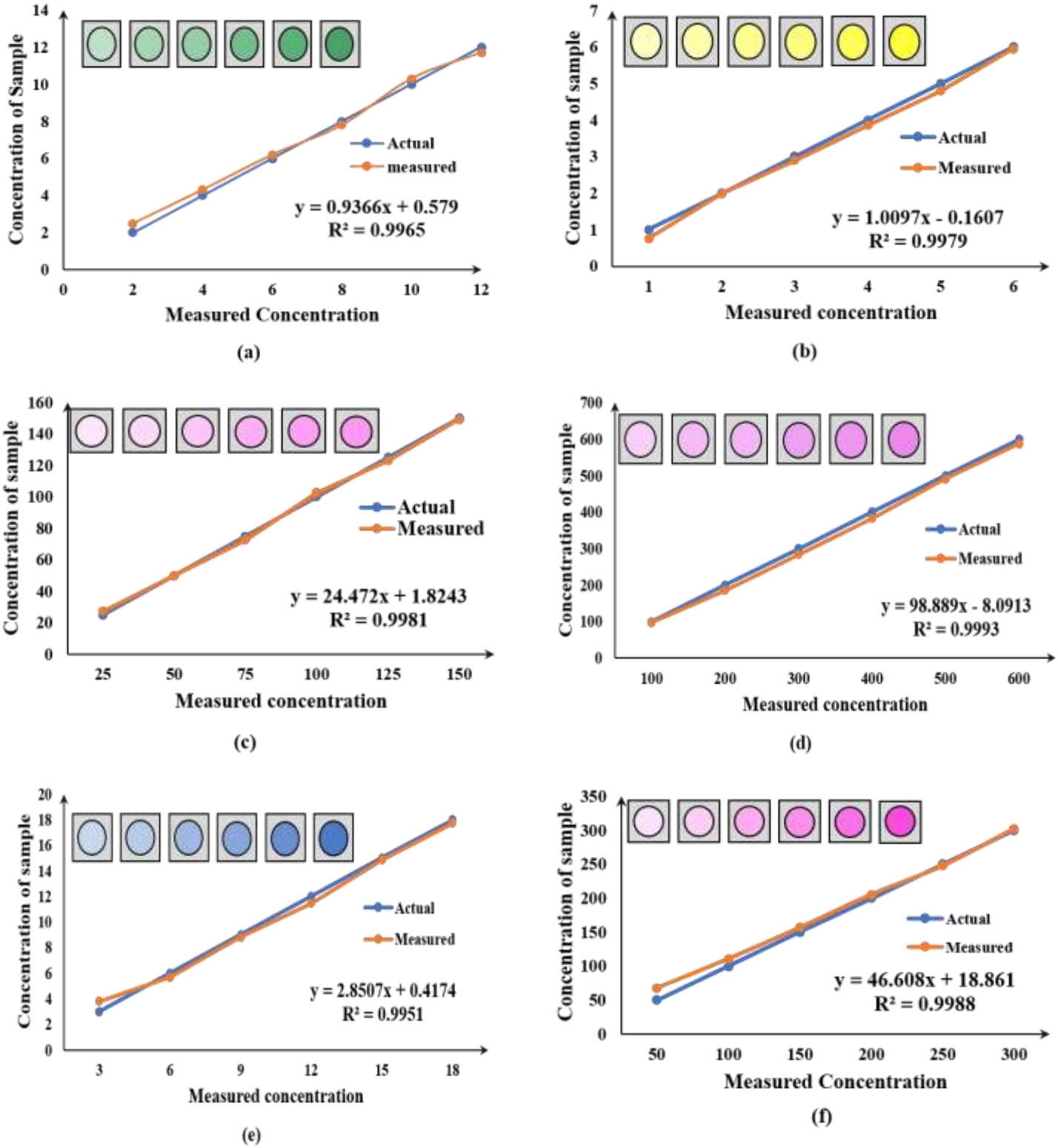


Fig. 6: Comparison between actual concentration of sample and measured concentration of sample by paper-based portable biochemistry analyzer (a) albumin, (b) Creatinine, (c) Urea, (d) Triglyceride, (e) Total Protein, (f) glucose

Fig. 5(a) illustrates the comparison between actual and measured albumin levels. Both datasets exhibit a strong positive correlation, with measured values slightly higher than actual values at certain points. The consistent upward trend suggests that the measurement method is reliable, closely approximating the actual albumin levels with minor deviations [16]. The limit of detection (LOD) for the measurement was 0.134 g/dL, with a coefficient of variation (CV) of 0.963%.

Fig. 5(b) shows creatinine concentration over time, with both actual and measured values increasing linearly. The measured values are slightly lower than the actual ones, indicating a small underestimation. The spectral data at 30 and 90 seconds correlates with this upward trend, reflecting a consistent rise in concentration. The limit of detection (LOD) for the measurement was 0.087 mg/dL, with a coefficient of variation (CV) of 1.311%.

Fig. 5(c) compares actual and measured urea concentrations at spectral values taken at 30 seconds and 90 seconds. Both datasets exhibit a strong linear correlation, with measured values closely aligning with actual concentrations across the range. The data suggests a consistent and reliable measurement method for urea, with minimal deviation between actual and measured values. The close overlap of the lines confirms that the spectral analysis effectively captures urea concentrations within the examined time intervals. The limit of detection (LOD) for the measurement was 0.122 mg/dL, with a coefficient of variation (CV) of 1.772%.

Fig. 5(d) compares actual and measured triglyceride levels across various values. Both the actual and measured data follow a nearly identical upward trend, indicating a strong correlation. The values increase proportionally, suggesting the measurement method is accurate and closely aligns with actual triglyceride levels. The limit of detection (LOD) for the measurement was 0.055 mg/dL, with a coefficient of variation (CV) of 0.812%.

Fig. 5(e) compares actual and measured total protein levels across different concentrations. The measured values closely follow the actual values, with minor deviations. Both lines show a steady, positive correlation, indicating that the measurement method accurately tracks protein concentration within a small margin of error. The limit of detection (LOD) for the measurement was 0.094 g/dL, with a coefficient of variation (CV) of 1.334%.

Fig. 5(f) presents a comparison between the actual and measured glucose concentrations, showing a noticeable difference at lower glucose levels (50–100 mg/dL). As the concentration rises, the two lines converge, indicating that the analyzer achieves greater accuracy at higher glucose levels (around 300 mg/dL). Given the minimal error between the actual and measured values, these discrepancies can be considered negligible. The limit of detection (LOD) for the measurement was 0.015 mg/dL, with a coefficient of variation (CV) of 0.212%.

In conclusion, the paper-based portable biochemistry analyzer is a reliable and accurate tool for real-time biochemical analysis. It consistently demonstrates strong alignment between measured and actual values for various analytes, including glucose, total protein, triglycerides, albumin, urea, and creatinine, with minimal error. The device also proves its reliability through spectral measurements taken at different intervals (e.g., 30, 90, 150 seconds), further validating its utility. This makes it an effective solution for point-of-care diagnostics, especially in remote or resource-limited areas where traditional lab equipment may not be available.

D. Repeatability Analysis

Repeatability test ensures that the device produces consistent results under the same conditions, highlighting its precision and reliability. This is vital in clinical applications where reliable data is necessary for diagnostics. Repeatability also validates the design and functionality of the analyzer, confirming that it performs effectively in various trials. The testing was conducted for various analytes, including albumin, creatinine, urea, triglyceride, total protein, and glucose, at their respective concentrations. The same gain rate of the sensor was set to ensure uniform conditions. For albumin, a sample concentration of 4 mM was used. Creatinine was measured using a concentration of 2 mM. For urea, a 50 mM sample was used. Triglyceride was analyzed using a 200 mM sample, total protein with a 6 mM concentration, and glucose was measured with a concentration of 100 mM. Five trials were conducted for each analyte, and the results were obtained to assess the precision and repeatability of the paper-based portable biochemistry analyzer, ensuring reliable and consistent data [16], [17].

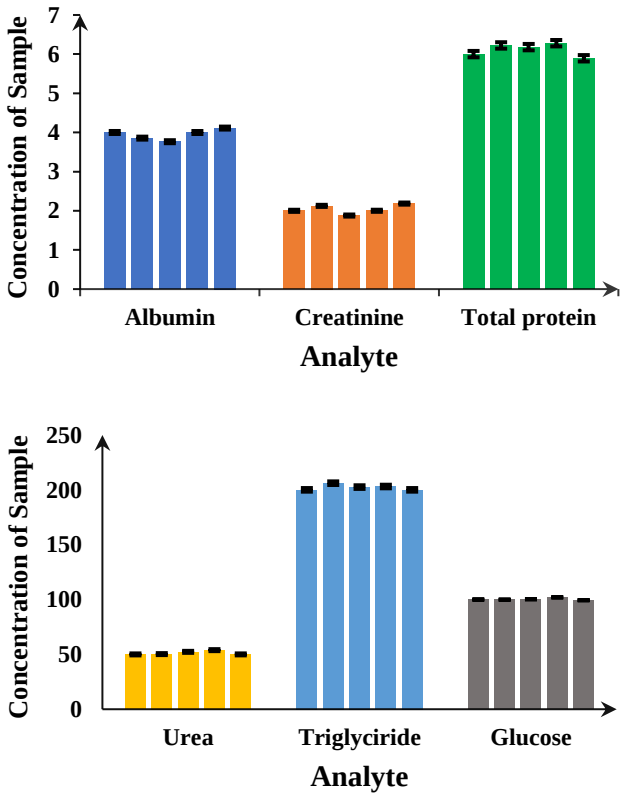


Fig. 7: A repeatability test was conducted for all analytes using the paper-based portable biochemistry analyzer, Error bars represent the standard deviation across five trials, indicating minimal variation between measurements.

The analysis demonstrates that the results for all analytes—albumin, creatinine, urea, triglyceride, total protein, and glucose—remain constant under identical conditions, as clearly

shown in the graph. Minimal standard deviation was observed across the five trials for each analyte, confirming the precision and repeatability of the paper-based portable biochemistry analyzer. Standard deviation is crucial in this context, as it measures the extent of variation from the mean, indicating the consistency of the results. A low standard deviation, as observed here, suggests that the readings are tightly clustered around the average, ensuring the reliability and accuracy of the analyzer for clinical or point-of-care applications.

IV. CONCLUSION

The portable, paper-based biochemistry analyzer is a breakthrough in point-of-care diagnostics, particularly for low-resource and remote healthcare settings. Utilizing wax-printed microchannels (µpads) coated with specific reagents, the device accurately detects critical blood parameters like albumin, creatinine, glucose, and urea. A multi-spectral sensor, interfaced with an ESP32 microcontroller, measures light absorption in real-time to provide precise concentration analysis. The device demonstrates excellent reliability, with correlation coefficients above 0.98 for all analytes and repeatability tests showing a standard deviation of less than 2%. The µpads, optimized to use only 100 µL of sample, ensure efficient and practical testing.

Compact, affordable, and user-friendly, the analyzer addresses the limitations of traditional lab equipment, which can be bulky and expensive, especially in underserved regions. Its innovative design offers a convenient alternative for rapid and accurate diagnostics, contributing to improved healthcare accessibility. The developed biochemistry analyzer represents a significant advancement in diagnostics, with the potential to enhance healthcare delivery in areas where conventional solutions are impractical, ultimately improving patient outcomes through timely and precise health assessments.

V. ACKNOWLEDGEMENT

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